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Tableau: Design Choices & Key Insights

The reason behind using a map for the 01_global terrorism heatmap was that the terrorism data has an inherent geographical nature. Hence, distribution of terrorist attacks on a map according to the country can make regional patterns and hotspots easy to recognize. I used color gradient to differentiate between the intensity of attacks according to the number of incidents taking place in that country. Finally, I added the year filter to allow for interaction according to the time period chosen by the user.

For the visualization of 02_top 10 terrorist groups over time, I used the stacked area chart to see the changing activity patterns of the terrorist groups over time. Using this graph, the activity pattern of individual terrorist groups as well as the changing trends over time can be seen very easily. I used the gname field to represent the various groups with a different color code while putting the iyear on the x-axis. Also, including the legend makes it easy for viewers to identify which group belongs to what color and thus analyze their activity pattern.

I employed a scatterplot in visualizing 03_Attack types and casualties due to its capability of representing the relation between two quantitative variables in the dataset which, in this case, is the number of deaths and wounded people in every incident. Coloring by attacktype1_txt enables the user to distinguish how different attack methods vary in terms of severity. Point size on the other hand allows for visualization of the most severe incidents. Finally, I added the functionality of hover details on every point so the user can inspect any incident without cluttering the graph with text.

For the 04_combined dashboard in Tableau, I created a combination of a line chart, bar chart, and a table to be used to visualize the monthly incidents, attack types, and top 5 attacked countries, all in one screen view. Time trends can therefore be easily viewed alongside methods and geographic distribution of terrorist attacks. Cross filtering is also supported whereby users are able to click on an attack type or a particular country in order to have the other visualizations updated with information from that point. Through this dashboard, one can tell that although there may be occurrences of terrorist acts throughout the whole year, certain months will tend to witness more attacks compared to others, implying a moderately seasonal trend of terror incidents. According to the data, months May and October experienced the highest terrorist attacks (with 1,079 and

1,018 recorded incidents respectively) compared to other months, with June being the month when the lowest number of attacks (703) was witnessed. It is clear from the bar chart above that Armed Assaults and Bombings/Explosions are the two most common attack types compared to other tactics such as hijackings. The top countries attacked table indicates that a few countries such as El Salvador, Guatemala and Nicaragua account for a disproportionately large share of incidents in this region, suggesting that terrorism risk is highly concentrated geographically.

Power BI: Design Choices & Key Insights

For 01_Geospatial Analysis of Attacks, I utilized a map visualization due to the geographical nature of the data, which shows the relationship between terrorist attacks and location. The visualization helps the user see which regions had the highest number of attacks by plotting the coordinates (latitude and longitude). By using a bubble size to represent the number of casualties, the user can easily compare attack locations and severity. In addition, slicers have been added to enable the user to select the year and region of interest.

For 02_Top 5 Terrorist Groups Comparison Visualization, a multi-row card visualization has been used since it enables the user to compare several metrics for each group at once in an easy way. It is necessary to include total attacks, casualties, and success rate in the metric list to give a better understanding of the terrorist activity of the group and not just its frequency. In addition, a slicer has been added to allow users to filter results by year.

The treemap was chosen for 03_Weapons Type Analysis as it helps to compare the number of different weapon usage frequencies within a small area. The size of the blocks reflects the total number of the attacks made while the intensity of the colors allows us to determine which category is more lethal among the presented categories. There is also the possibility to conduct the analysis starting with general categories and then proceeding to sub-categories of different weapons used.

I designed the 04_target analysis dashboard in such a way that it could incorporate a donut chart, clustered column chart and detail table to provide an opportunity to analyze target patterns in different levels of details on a single screen. The donut chart illustrates general statistics concerning the types of targets while the clustered bar chart presents data regarding casualty figures in different target categories. Lastly, the detail table emphasizes the most targeted targets. From the dashboard, it was observed that there are only a few types of targets where most of the attacks occur, with Military and Private Citizens & Property being the most common among these targets both in number and in

terms of the effect. As seen from the casualty graph, some targets suffer a greater number of casualties compared to other targets; thus, it can be concluded that all frequently attacked targets do not have an equal impact on human lives.

Tableau: Interaction steps

1. Use the Year filter on the heatmap to change the time period and see how country level attack intensity changes.
2. Hover over any country, terrorist group area, or scatter-plot point to view detailed tooltips with incident counts and casualties.
3. In the Top 10 Terrorist Groups area chart, use the gname filter (or legend) to focus on specific groups over time.
4. On the combined dashboard, click a bar or country row to cross-filter the other charts, then click again (or press Esc) to clear the selection and show all data.

Power BI: Interaction steps

1. On the Geospatial Analysis page, use the year and region slicers to filter the map; hover over bubbles to see the exact location and casualty details.
2. On the Terrorist Group Comparison page, use the year slicer to update the multi-row cards, and scroll to view all selected groups.
3. On the Weapons Type Analysis page, click any treemap block to filter to a specific weapon type; use drill-down (double-click or the drill icon) to see weapon subtypes.
4. On the Target Analysis Dashboard, click donut slices or bars to filter by target type, and use the bookmark buttons (Overview / Detailed Targets) to switch between the summary charts and the detailed targets table.